

NORRKOPING, Sweden

WMO: 025740

Lat: 58.583N

Lon: 16.147E

Elev: 42

StdP: 100.83

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.5	-12.1	-17.6	0.8	-15.3	-14.5	1.1	-10.8	7.3	3.2	6.5	3.3	0.7	260	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	27.3	18.2	25.5	17.7	23.7	16.9	19.8	24.4	18.9	23.1	17.9	22.2	2.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.4	21.3	17.3	12.4	20.5	16.2	11.5	19.9	56.9	24.6	53.7	23.0	50.6	22.3	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.6	5.0	-17.4	29.9	4.6	1.9	-20.7	31.3	-23.4	32.4	-26.0	33.4	-29.3	34.8		
(5)			-17.9	21.2	4.4	1.3	-21.1	22.1	-23.7	22.9	-26.2	23.6	-29.4	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-1.7	-1.6	1.3	6.3	11.4	15.2	18.1	16.9	12.9	7.4	3.3	0.3
	DBStd	7.88	4.83	4.61	4.35	3.36	3.33	2.82	2.72	2.52	2.96	3.52	3.66	4.89
	HDD10.0	1707	363	326	269	118	23	1	0	0	10	95	201	301
	HDD18.3	3996	621	559	527	361	217	101	39	58	164	341	450	559
	CDD10.0	804	0	0	1	8	65	157	250	214	96	13	1	0
	CDD18.3	52	0	0	0	0	1	6	31	14	1	0	0	0
	CDH23.3	451	0	0	0	1	17	78	259	93	2	0	0	0
	CDH26.7	79	0	0	0	0	1	12	54	13	0	0	0	0
Wind	WSAvg	2.3	2.4	2.4	2.6	2.5	2.3	2.3	2.1	2.0	2.2	2.3	2.5	2.6
Precipitation	PrecAvg	611	43	36	30	35	42	68	73	72	58	55	52	48
	PrecMax	795	86	60	57	88	89	125	141	174	129	140	117	88
	PrecMin	438	8	14	8	3	8	13	5	9	11	3	11	13
	PrecStd	73	20	12	14	22	20	27	34	34	33	29	25	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	8.9	14.4	20.5	25.5	28.4	30.4	28.1	23.4	17.2	12.6	11.2
		MCWB	6.9	6.2	8.5	11.7	16.1	17.8	19.3	18.6	16.7	13.6	11.1	9.2
	2%	DB	6.8	6.9	11.5	17.5	22.9	25.7	28.1	25.7	21.4	15.1	10.8	9.0
		MCWB	5.4	4.9	6.8	10.3	14.4	16.9	18.6	18.3	15.7	12.6	9.5	7.4
	5%	DB	5.5	5.5	9.2	14.9	20.5	23.4	26.2	23.8	19.5	13.7	9.6	7.5
		MCWB	4.2	3.9	5.7	9.1	13.4	15.6	18.1	17.6	14.8	11.7	8.4	6.1
	10%	DB	4.2	4.2	7.3	12.7	18.2	21.5	24.3	22.2	17.9	12.5	8.4	6.3
		MCWB	3.0	2.7	4.6	7.6	12.1	14.8	17.6	16.6	14.2	10.7	7.3	5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	6.6	9.1	13.1	17.4	19.3	21.4	21.1	17.8	14.8	11.4	9.3
		MCDB	8.3	8.2	13.5	19.2	23.1	24.9	26.2	25.3	21.3	16.5	12.3	10.8
	2%	WB	5.6	5.2	7.2	11.0	15.7	17.8	20.1	19.6	16.4	13.1	9.6	7.5
		MCDB	6.6	6.5	10.6	16.0	20.6	23.9	25.0	23.3	19.9	14.6	10.6	8.7
	5%	WB	4.3	4.1	6.1	9.5	14.2	16.6	19.3	18.5	15.5	12.0	8.5	6.2
		MCDB	5.4	5.4	8.7	14.2	18.9	21.9	23.9	22.0	18.5	13.4	9.4	7.3
	10%	WB	3.1	2.9	4.9	8.1	12.9	15.5	18.4	17.5	14.6	11.0	7.4	5.1
		MCDB	4.0	4.0	6.9	11.8	17.3	20.2	22.9	20.9	17.2	12.4	8.2	6.1
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.4	7.4	9.6	10.0	9.4	9.2	8.8	7.9	6.4	4.3	4.4
		MCDBR	4.9	5.7	10.0	13.9	14.3	13.6	12.7	11.5	10.5	7.5	5.4	4.8
		MCWBR	4.5	4.3	6.3	7.8	7.1	6.1	4.9	5.2	5.7	5.3	4.6	4.5
	5% WB	MCDBR	4.8	5.2	9.1	12.3	12.1	11.7	10.3	9.6	9.7	6.8	5.3	4.7
		MCWBR	4.6	4.2	6.1	7.4	6.7	5.9	4.8	4.9	5.6	5.3	4.7	4.5
Clear-Sky Solar Irradiance	taub	0.284	0.321	0.340	0.378	0.374	0.362	0.389	0.380	0.348	0.338	0.310	0.270	
	taud	2.243	2.297	2.351	2.309	2.386	2.441	2.387	2.409	2.484	2.419	2.290	2.177	
	Ebn at Noon	545	701	803	827	859	874	838	818	793	679	505	425	
	Edh at Noon	46	70	88	108	107	103	107	98	78	63	44	35	
All-Sky Solar Radiation	RadAvg	0.34	0.93	2.44	3.98	5.24	5.81	5.27	4.19	2.81	1.26	0.44	0.23	
	RadStd	0.05	0.16	0.25	0.41	0.43	0.27	0.52	0.40	0.35	0.20	0.09	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	-129	-158	N/A	N/A

Nomenclature: See separate page